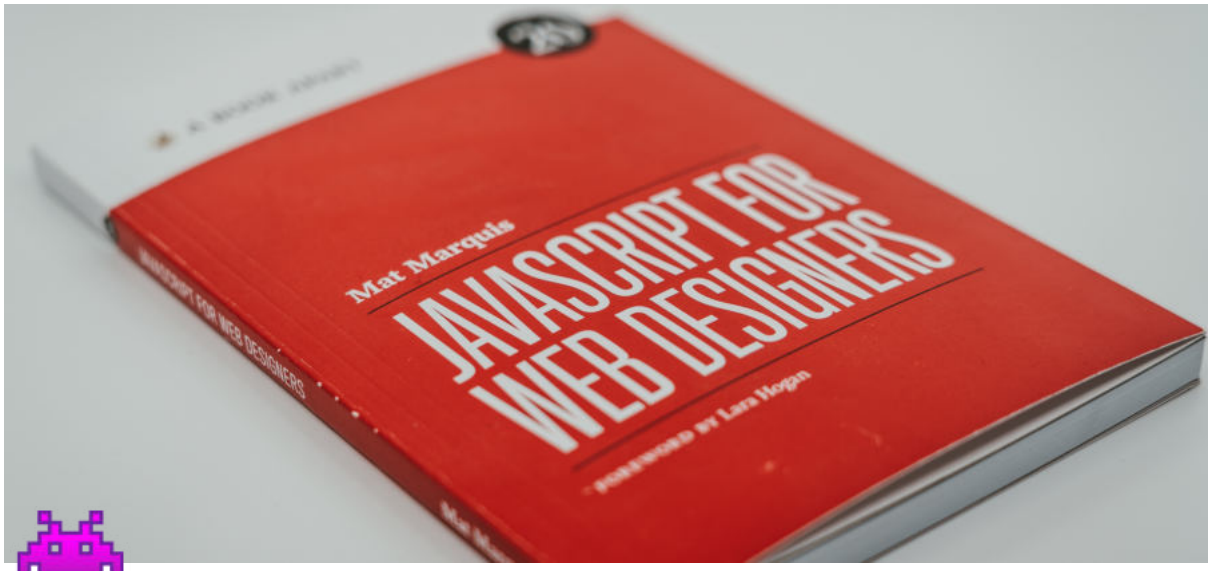


Designing APIs

Created	@February 13, 2022 11:12 PM
Type	Lecture
Week	8
Lecture #	1
Lecture URL	https://youtu.be/9IM2zpiWQ7A
Notion URL	https://21f1003586.notion.site/Designing-APIs-I-1e8ec3628dac4601b70cbb3797026197

What makes a good API?

- There is no strict consensus
- A ton of conflicting opinions

Review of ideas from https://cloud.google.com/apigee

Web API Design: The Missing Link

- eBook from Apigee: useful guidelines

Purpose of an API

- To be used to build applications
- Design APIs with developers in mind

Remote Procedure Call → used to connect to a remote server and retrieve data based on some function arguments

Data oriented approach

Entities

- Students
- Courses
- Grades

Actions

- Add, Edit, Delete

Summaries

- List
- Grade point average
- Top students in courses
- ...

Exotic

- Students with names beginning with ‘A’ who score more than 85% in MAD1

Possibilities

- List of students
  - http://localhost/getListOfStudents
- Individual student
  - /getStudent?id=xyz
- Add new student
  - /createNewStudent
- Edit existing student

    - /editStudent?id=xyz

Can get pretty complex

- /getTopStudents
- /createStudentAndAddToCourse

Not fundamentally wrong

- Difficult to remember
- Challenging to document, understand

Use conventions

- List of students
    - http://localhost/students
- Individual student
    - http://localhost/student/123
- Add new student
    - POST .../student
- Edit existing student
    - PATCH .../student/123

URL conventions

- Nouns in the URL are good, verbs are bad
    - /student
    - PATCH /student/123

  instead of

    - /create/student
    - /edit/student/123
- Verbs in HTTP → use the Method
- well-known URLs
    - Can API be “discovered” by crawling from `/` ?
    - Standard conventions for listing, posting, etc.

- Permalinks
  - not necessarily human readable
  - unique ID for posts, documents, etc

Query URLs

- /search?course=123&type=student

OR

- /course/123/students

Convention → structured URLs preferred by developers

Why not /course-123-students?

- Can also be used — either way the URL path has to be parsed
- Same complexity — just difference in developer experience

Lecture #2 starts here

Lecture URL: https://youtu.be/VM-2WK2KG4I

HTTP Verbs

- GET
  - read data, lists etc
  - cacheable — all data is in the URL
- POST
  - create new object/data
  - Not cacheable in general — data not part of the cache index
  - Can be used for reading data
- PUT / PATCH
  - Update with all the new data / incremental new data → PATCH to be preferred

These are all conventions

Output formats

- Structured data
  - XML → very good
  - JSON → not so good — very limited data types
- Simplicity
  - JSON → Human readable
  - Easy to parse — but can have problems
- JSON + extensions is preferred format at present
  - Not necessarily the best possible

Included links

- Just seeing JSON output, cannot know the query
- Include additional links that give pointers to other useful information
  - Could have been done as part of documentation
  - In some cases more flexible to provide with response

Example → GitHub API

API URL → https://api.github.com/users/kashifulhaque

Response


```
{
  "login": "kashifulhaque",
  "id": 37375667,
  "node_id": "MDQ6VXNlcjM3Mzc1NjY3",
  "avatar_url": "https://avatars.githubusercontent.com/u/37375667?v=4",
  "gravatar_id": "",
  "url": "https://api.github.com/users/kashifulhaque",
  "html_url": "https://github.com/kashifulhaque",
  "followers_url": "https://api.github.com/users/kashifulhaque/followers",
  "following_url": "https://api.github.com/users/kashifulhaque/following{/other_user}",
  "gists_url": "https://api.github.com/users/kashifulhaque/gists{/gist_id}",
  "starred_url": "https://api.github.com/users/kashifulhaque/starred{/owner}{/repo}",
  "subscriptions_url": "https://api.github.com/users/kashifulhaque/subscriptions",
  "organizations_url": "https://api.github.com/users/kashifulhaque/orgs",
  "repos_url": "https://api.github.com/users/kashifulhaque/repos",
  "events_url": "https://api.github.com/users/kashifulhaque/events{/privacy}",
  "received_events_url": "https://api.github.com/users/kashifulhaque/received_events",
  "type": "User",
  "site_admin": false,
  "name": "Kashif",
  "company": null,
```

```
  "blog": "https://ifkash.pages.dev",
  "location": "Kolkata, India",
  "email": null,
  "hireable": true,
  "bio": "Data Science undergrad @ IIT Madras • I occasionally contribute to @k2-labs ",
  "twitter_username": "notifkash",
  "public_repos": 44,
  "public_gists": 4,
  "followers": 24,
  "following": 83,
  "created_at": "2018-03-14T15:07:06Z",
  "updated_at": "2022-02-05T13:07:11Z"
}
```


Authentication

- Token based authentication
- OAuth2 → Good standard used in many places
- JWT (JSON Web Tokens) and other variants possible

Use standard techniques wherever possible

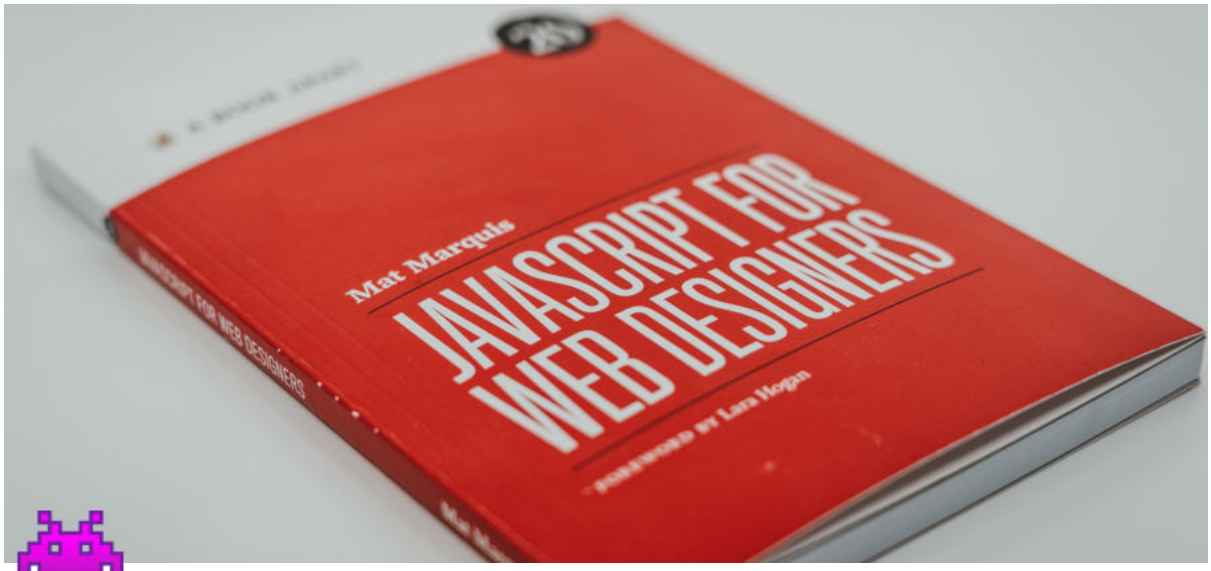
Summary

- Good API design requires experience
- Mostly based on conventions → no rigid rules
- But conventions are important

APIs are designed for developers, not end users

Problems with REST

- Most RESTful APIs are violating some constraint of REST
- REST is an architecture style → Not an API design document
  - Not rigid guidelines — sometimes bending the rules can help
- Chatty — Multiple requests to fetch data for a view
  - First get details of student, the list of courses taken by the student, then details for each course, then aggregate marks in each course ...
- Specific requests permitted — not a general “Query Langauge”
- Cannot specify “what is needed” — need to break up into individual requests then get the results



Intro to GraphQL

Created	@February 13, 2022 11:42 PM
Type	Lecture
Week	8
Lecture #	3
Lecture URL	https://youtu.be/x4eZPjzM92Q
Notion URL	https://21f1003586.notion.site/Intro-to-GraphQL-8e4d3ada11d741df8054e6d102f2066c

Why GraphQL

- REST based APIs are endpoint based
  - Specific types of queries permitted
  - Complex data requests must be constructed with multiple GETs
  - /student?name='A'&age='It_25'
    - Special characters? arbitrary queries?
- Multiple data sources

    - Modern sites require inputs from multiple sources
    - Simultaneous query and fusion of data — at client or at server?
- Declarative programming — what to do, not how to do it
    - Very useful in view construction
    - Improves developer experience
    - Why not for retrieving data as well?

How GraphQL?

How to GraphQL: https://www.howtographql.com/

- Engine on the server side to handle requests
- Translate requests in a complex query language to data requests
    - Collect data, filter etc. on the server
    - Respond to the client only with the data needed

What is GraphQL?

- Query language
- Can be used over HTTP
    - Usually with POST
- Send complex queries over POST body

Layer between client and server

- Receives complex queries
- Convert to (multiple) queries to the server, fuse results

Type system

- Specify types of query items
    - String, Int, Collection of items, etc
    - Automatically catch and prevent certain query errors
- Specify relations between items
    - Student → [Course]: student can have list of course

API versioning: Evolve

- Requests are JSON-like
- Add functionality as required
- Deprecate functionality if neede
- Not necessary to define new API versions in most cases

Mutation

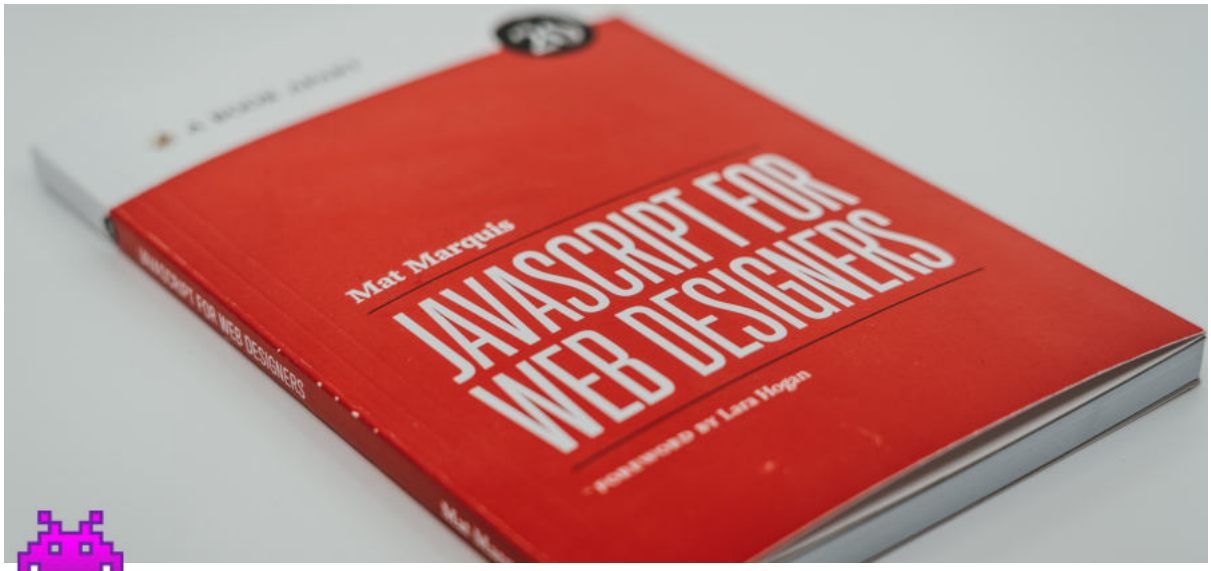
- Create/Update/Delete type operations
- Generalized to be any kind of query
  - Alter the underlying data store

Tools

- Apollo server → system to build up GraphQL
  - Connect to multiple backends
  - Define own resolves
- Explorers
  - Google, github, graphql.org
  - dynamically construct and test queries

Summary

- Extension of core API concepts
- Integrate with multiple data sources
- Complex query language
  - Filter data at the server end, send only relevant data to the client
- Does not necessarily reduce server complexity
  - Server may even become more complex
  - But for complex queries this might have been needed anyway



Markup Alternatives

Created	@February 13, 2022 11:54 PM
Type	Lecture
Week	8
Lecture #	4
Lecture URL	https://youtu.be/Po4hmc3-vH4
Notion URL	https://21f1003586.notion.site/Markup-Alternatives-81e50c1d2d594abaa6faa4f0913bed4b

Why HTML?

- General enough markup for all text
- “Living standard”
  - Can adapt to many new functional requirements
- Extensible
  - WebComponents, JS enable new tags if needed

Focus on “semantic” content → leave styling to CSS

Why not HTML?

- Structured data communication
  - JSON often used, but not really designed for this
  - XML etc. much better, but overkill
- Virtual Reality, new environments
  - VRML
  - X3D
- Still relatively verbose for humans

Text-based markup

- Write almost like normal text
- Use inline markers: **strong**, _emphasised_, # HEADING
- Many alternatives
  - Markdown
  - ReStructured Text (RST) → documentations
  - AsciiDoc

Markdown Example

```
# This is a heading

And this is a regular paragraph

* A bullet list
* Another bullet list

1. A numbered list
2. More numbers

And
[links](https://www.example.com)
```

Why text?

- Uniform character representations agreed upon - ASCII, Unicode
  - Write once, read anywhere
- Guard against obsolescence

    - Old file formats not readable
    - Not easy way to reverse engineer
- Compact
- Easy for humans, also to read

Why not text?

- Hard to encode “structure”
- Ambiguity possible
    - Parsing the format may not be unique
- More focused towards English and Roman alphabet
    - Possible to create reasonable equivalents in other languages/scripts

Compile/Convert

- Systematic conversion between markup formats
    - Just like any other language <_> language conversion
- Easier between structured languages
    - XML - SGML etc
- Custom compilers

Pandoc: “Swiss Army Knife” tool to convert between formats

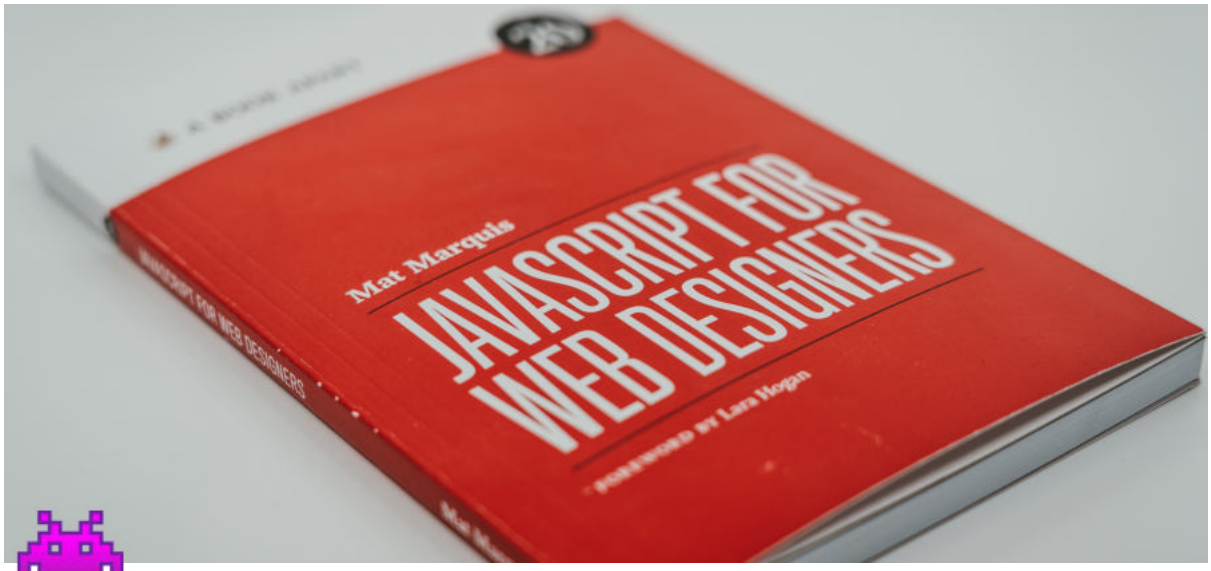
Mixed functionality

- Programs mixed with documentation
    - Web/Weave, Doxygen with similar comment-oriented systems
- JSX, Vue
    - Mix JS with templates and HTML structure

Summary

- Markup design has lots of constraints
    - Structured data
- Focus more on human interface

    - Understandable markup
    - Easy to write and transmit
- Compilers/Converters to handle interchange



JAM Approach

Created	@February 14, 2022 12:09 AM
Type	Lecture
Week	8
Lecture #	5
Lecture URL	https://youtu.be/YrcQ_aD3rYg
Notion URL	https://21f1003586.notion.site/JAM-Approach-4dc672223eac4cdb84587a164c43e3cb

JAM

- JavaScript
- APIs
- Markup

What does an app need?

- Data store → What the app is for

    - Access and retrieval → APIs
    - SQL, NoSQL, GraphQL
- User Interface → to interact with the user
    - Vanilla HTML + forms → request/response
    - JavaScript → interactivity, closer to native
- Business logic → what should be done with the data
    - Backend computation → Python, Go, Node.js
    - Frontend computation → JavaScript, WebAssembly (WASM)

Content Management Systems

Example → Blog application

- CRUD for posts, comments
- Ratings for posts, comments
- User management
- Analytics

All these are data manipulation → can be independent of the UI

Wordpress

- One of the oldest and most popular
- Handles both data storage (backend) and templating (frontend)
- Also provides API → https://developer.wordpress.org/rest-api/

API can be used to build a CMS without frontend

Static Site Generators

- NextJS, NuxtJS, Gatsby
    - JS based → useful for interactive sites, complex designs, plugins
- Jekyll, Hugo
    - Primarily text oriented
    - Blogs, homepages

Why Static Site Generators (SSGs)?

- Servers can focus on delivering content
  - Static files faster to fetch
  - “compile time” optimization to reduce the file transfer

“First Contentful Paint”

- Pure HTML allows easy transfer and parsing, can be displayed quickly

JS Hydration

- Static HTML transferred from server — no interactivity
- “Hydrate” the HTML with event handlers
  - Inject interactivity after initial rendering is complete
- Delayed, but still fast enough
  - Good combination of speed and interactivity

JAMStack → pinnacle of web app development?

- Takes care of storage + logic + presentation type apps
  - APIs flexible enough to handle any backend
  - Markup easy to change or compile
  - JS powerful enough to emulate any other behaviour
- Other developments?
  - Real-time communications
  - New interface devices, displays
- JAM approach general enough to extend with APIs
  - Until hit by performance issues
  - Wait for the Next Big Thing